

Operating Systems — Week 1 Q&A

Q1. What is paging? [5 marks]

Ans: Paging splits virtual memory into fixed-size pages mapped to physical frames via a page table. The TLB caches recent translations to avoid repeated page-table walks.

Question 2: Define context switch. [10 marks]

Answer: The act of saving one process's CPU state and loading another's. It includes register saves, TLB flush on some architectures, and updating the kernel's run queues.

Q3. Explain round-robin scheduling. [15 marks]

Ans: RR assigns each ready process a fixed time quantum. Small quanta improve responsiveness but increase context-switch overhead. Large quanta degenerate toward FCFS.